Yului Jeong

Undergraduate Researcher | Seoul National University
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IMSI Lab profile

Research Focus

Earth observation AI · multi-temporal SAR flood segmentation · hyperspectral imaging · energy-efficient spiking neural networks · medical image analysis

Education

Seoul National University — B.S. in Earth and Environmental Sciences | *Mar 2024 – Feb 2028 (expected)*

Next-Generation Semiconductor Microdegree, COSS | *Aug 21, 2026*

- Completed Next-Generation Semiconductor (General) Program; non-degree credential.

Research Experience

IMSI Research Lab, Seoul National University — Undergraduate Researcher | *Feb 2026 – Present*

- Led first-author research manuscripts SARFI, STELA, and USARGI for IEEE ICCE 2027.
- Research across multi-temporal SAR flood mapping, 3D multiphase MRI, and hyperspectral remote sensing AI.

Satellite Geophysics Research Laboratory, Seoul National University — Research Project Participant | *2026*

- Developed STORM-SNN for energy-efficient multi-temporal Sentinel-1 SAR flood mapping.
- Research presentation accepted at the Korean Society of Remote Sensing (KSRS 2026).

Geomicrobiology Lab (Web Lab), Seoul National University — Undergraduate Researcher | *Mar 2025 – Jun 2026*

- Investigated environmental microorganisms and anaerobic cultivation for bioelectrochemical applications.
- Conducted geological field/laboratory research and mineralogical characterization using XRD, SEM–EDS, and FT-IR.

Precision Bioinstrumentation Lab — Undergraduate Researcher | *Sep 2025 – Mar 2026*

- Worked on automation of drug-treatment experimental workflows and cell-response imaging analysis.
- Independently developed a skeleton-based quantification method for linear biological specimens and filed a related Korean patent application.

Accepted Conference Papers

- *SpecRule: Automated Discovery of Interpretable Spectral Rules for Hyperspectral Classification* — IEEE ICCE-Asia 2026 · Accepted [First author]
- *SpecLite: Resource-Efficient Hyperspectral Classification with Fourfold Spectral Band Reduction* — IEEE ICCE-Asia 2026 · Accepted [First author]
- *SCINTILLA-SNN: A Spiking Multi-Scale Selective Aggregation Network for Perineural Invasion Prediction* — IEEE APCCAS 2026 · Accepted [Co-first Author] link
- *SpikeDS: Dual Sparsity Spikformer for Perineural Invasion Prediction in 3D MRI* — IEEE AICAS 2026 · Oral [Presenting author] link
- *Adaptive Routing for Efficient Diffusion Transformer-Based PNI Prediction* — IEEE AICAS 2026 · Oral
- *Anatomy-Privileged Distillation with Token Routing for MRI-Based Prediction of Perineural Invasion* — IEEE AICAS 2026 · Oral
- *Order-Aware 2.5D Multiple Instance Learning for Preoperative MRI-Based PNI Risk Assessment* — IEEE APCCAS 2026 · Accepted link
- *WSPolypNet: Weakly Supervised Polyp Localization in Colonoscopy Videos* — IEEE ICCE-Asia 2026 · Oral link

First-Author Manuscripts Under Review

- *SARFI: A Persistence- and Emergence-Aware Spiking Network for Multi-Temporal SAR Flood Segmentation* — IEEE ICCE 2027 · Under Review [First author]
- *STELA: Learning Patient-Specific State Transition Evidence for Perineural Invasion Prediction in Multiphase MRI* — IEEE ICCE 2027 · Under Review [First author]
- *USARGI — Guided Sampling for Rare-Class Flood Segmentation in Urban SAR Imagery* — IEEE ICCE 2027 · Under Review [First author]

Selected Presentations & Collaboration

IEEE AICAS 2026 · SpikeDS | *Sep 2026 · Vietnam*
- Oral presenter.

KSRS 2026 · STORM-SNN | *Oct 2026 · Busan*
- Conference presentation accepted.

IEEE APCCAS 2026 · SCINTILLA-SNN | *Oct 2026 · Fukuoka*
- Accepted paper.

IEEE ICCE-Asia 2026 · SpecRule, SpecLite | *Oct 2026 · Gangneung*
- Two accepted first-author papers.

KBS Disaster Media Center · Earth Observation AI research discussions | *Aug 2026*

NVIDIA Research Hong Kong Collaboration Meetings | *Sep–Oct 2026 · Multiple sessions*

Samsung Medical Center · recurring clinical research discussions | *2026 · Approximately four visits*

Honors & Awards

Silver Award (2nd Place), Global Track — Global College Startup Camp | *Feb 2026 · Tokyo, Japan*

Grand Prize (Dean's Award), 7th Advanced Scientific Instrumentation Student Research Program | *Jan 2026 · SNU*

Top Excellence Award, Sejong University Startup Item Competition | *Jul 2025*

Grand Prize, 2025 SNU The Beginning Ideathon (SNU Entrepreneurship Center × SNAAC) | *Jun 2025*

Patent Applications

Skeleton-based Per-Length Staining Intensity and Distribution for Biological-State Quantification | *Independently developed · Korean patent application filed*

Automated Earth Science Inquiry with Real-Time Simulation–Language Model Consistency Verification | *Korean patent application*

Leadership, Teaching & Outreach

Kia Inspiring Class — Green Changer, Environmental Student Ambassador | *May – Aug 2024*

SNU College of Natural Sciences Science Reporter, Ja:Mong 7th cohort | *Apr 2024 – Feb 2026*

Science & University-Life Content Creator · Instagram @growth_yule | *Oct 2026 · 5,708 followers · 208 posts*

Invited Student Panelist, SNU College of Natural Sciences Education Forum | *Nov 2025*
- Advocated career-development, entrepreneurship, and university–industry engagement improvements.

SNU College of Natural Sciences Science Camp — Teaching Assistant & Mentor (two occasions) | *Aug 2024; Aug 2025*
- 40 certified volunteering hours in 2025.

Gwangmun High School — Assistant Instructor | *Aug 2025*

Yeosu Happiness Education Support Center — Assistant Instructor, Academic & Career Planning Workshop | *Apr 2025*

Sesangeul Pumeun Education (세상을 품은 교육) — Deputy Leader, Learning Solutions Team | *Nov 2024 – Oct 2025*

PhiSphere (Plant–Microbial Fuel Cell Startup) — Microbial Research Lead | *May – Sep 2025*
- Selected for SNU Innovation Startup Lab support in 2025 (official program period: Jul–Dec 2025).
- Admitted to the SNU Campus Town incubator in the second half of 2025.

The University Newspaper — Contributing Columnist | *Sep 2025*

Selected Science Writing

2026.07.14 3 학년이 알려주는 자연대 학부생 인턴

2025.10.10 실험실 밖 진짜 과학, 전재규 학술캠프 2 일차 지질답사를 훔쳐보다!

2025.08.06 저학년에게도 열린 연구의 문, 자연대 학부생 연구 인턴십 체험기

2024.09.20 가수 방예담, 서울대 자연과학대학에서 버스킹!

Additional Academic Training

Student-Directed Research, Seoul National University | *Summer Session 2026 · Completed*

Technical Skills

Python, PyTorch, Git; spiking neural networks; 3D MRI; SAR and hyperspectral imaging; XRD; SEM–EDS; FT-IR; anaerobic microbial culture; cellular imaging and drug-treatment workflow automation.